

NAMAN YESHWANTH KUMAR

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SUMMARY

MS Computer Engineering (ASU, GPA 3.81, May 2026) who builds production agents on LangGraph and iterates them to reliability. Designed SiliconCrew — a 21-tool LangGraph ReAct agent with eval-driven self-correction, multi-provider orchestration, and a full MCP server — scoring **37/92 on CVDP** benchmarks. Sole-built Trezzit (trezzit.com), a production AI SaaS serving **600+ users** for 12+ months, iterating an LLM pipeline from **60% → 95%+** accuracy across 4 providers.

TECHNICAL SKILLS

Agent Architecture: LangGraph | LangChain | ReAct Agents | MCP Protocol | Eval & Self-Correction Pipelines | Multi-provider LLM Orchestration | RAG Pipelines | Tool Use / Function Calling | Pydantic Validation

Full-Stack: Python | TypeScript | FastAPI | Django REST Framework | React | Next.js | WebSocket | Async Python | PostgreSQL | Redis | Celery

Infra & Observability: Docker | GitHub Actions CI/CD | DigitalOcean | AWS | Nginx | Per-provider LLM Cost & Latency Tracking | Linux

EDUCATION

Master of Science — Computer Engineering (GPA 3.81/4.0)

Expected May 2026

Arizona State University

Tempe, AZ

Coursework: Artificial Intelligence · Reinforcement Learning in Robotics · Perception in Robotics · Foundations of Algorithms ·

Algorithm/Hardware Co-Design

Bachelor of Engineering — Electrical and Electronics

August 2022

BMS College of Engineering

Bengaluru, India

PROJECT EXPERIENCE

SiliconCrew — Production LangGraph Agent with Eval Framework & MCP Server

Nov 2025 – Present

- Built a 21-tool LangGraph ReAct agent with multi-provider orchestration (Gemini, OpenAI, Anthropic) and provider-specific quirk handling — chose LangGraph over linear chains for its state-machine graph structure, enabling conditional branching between generation, evaluation, and self-correction nodes; implemented a full MCP server (stdio/SSE/HTTP, auto-discovery of LangChain @tool functions) for Claude Desktop, VS Code, and any MCP client.
- Designed the eval-driven self-correction loop: on task failure the agent reads diagnostic output, identifies root cause, patches its generation, and re-verifies without human input — mandatory 3-iteration policy drives correctness where single-shot generation fails; achieved **37/92 first-pass** and **5/5 perfect score** on structured code generation benchmarks; a generalizable eval pattern applicable to any agent domain.
- Built end-to-end agent observability: real-time WebSocket streaming of tool-call execution traces, per-session token + cost tracking across all LLM providers enabling compute-aware model selection, and pipeline-stage artifact inspection — making agent reasoning transparent and debuggable without re-running jobs; FastAPI backend, Next.js + TypeScript frontend.

Trezzit — Production AI SaaS (trezzit.com)

Feb 2025 – Present

- Sole-built and shipped Trezzit (trezzit.com, **600+ users**, 12+ months live) — iterated receipt parsing from ~60% to **95%+** by diagnosing specific LLM failure modes (hallucinated item counts, currency ambiguity, multi-page layouts) and building a two-pass eval pipeline: extract structured data → semantically validate → escalate to a stronger model on failure; provider-abstracted registry across Gemini, OpenAI, Claude, and Grok.
- Production monitoring tracks per-provider LLM cost and latency used to make ongoing model selection decisions; **995 tests** on every push via GitHub Actions CI with live PostgreSQL; **60+ REST endpoints**, **40 Django models**, **341 React components**; Redis, Celery, DigitalOcean VPS.

HEAL.AI — Healthcare AI with Custom RAG Pipeline

Sep – Nov 2025

- Won 1st place at Devlabs Hackathon; built a custom RAG pipeline without an external vector database — PDF chunking with 2-sentence overlap, Google text-embedding-004 (768-dim) embeddings, cosine similarity retrieval with confidence scoring, and automated FDCPA-compliant dispute email generation via Gemini; FastAPI backend, React + TypeScript frontend.

Intel Automated Self-Checkout — Open Source Contribution

Jan – Feb 2025

- Contributed a real-time sensor data pipeline to **Intel's official open-source self-checkout reference architecture** — multi-transport ingestion layer (MQTT, Kafka, HTTP) for LiDAR and weight sensor streams, Grafana dashboard, and integration documentation; navigated an external production codebase, incorporated reviewer feedback, and had **PR #655 accepted by Intel engineers**.

WORK EXPERIENCE

Software Developer (Sole Engineer)

Oct 2022 – Apr 2024

NCR GOLD

Bengaluru, India

- Automated the entire order lifecycle for a wholesale business — production Django platform (PostgreSQL + Celery + Nginx) processing **18,000+ orders over 3 years**; reverse-engineered third-party ERP (zero public API) via SQL schema analysis for bidirectional sync, eliminating ~20 min of manual relay per order.

AWARDS & HONORS

- Won **1st place at Devlabs Hackathon** — HEAL.AI, an AI healthcare assistant with a custom RAG pipeline for insurance Q&A and automated medical bill dispute generation.
- Won **'Best Use of AI'** at Strategy X DevHacks'25 (ASU) — built AdaptED AI, a Gemini-powered platform generating personalized flashcards, quizzes, and learning paths from uploaded documents in a 24-hour hackathon.
- Secured **2nd place at Hack SoDA 2024** (Amazon-sponsored, 24-hour) — built PassGen, a secure offline Chrome Extension for password generation.